EN3150 Assignment 03

Resource-Constrained CNN for Edge Image Classification

EuroSAT RGB: accuracy, storage and computation

Group: **54**

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Final report — 17 October 2026

Based on saved experiment artifacts; no new full-dataset training was performed during this audit.

The CompactEuroSAT controlled experiment remains blocked by absent images.

Repository: `https://github.com/Abdul-Rahman-bme/EN3150-Group-Assignment-03-CNN`

GitHub profile: `https://github.com/Abdul-Rahman-bme`

Abstract

This study compares a standard convolutional neural network, a custom depthwise-separable network, an additional residual depthwise CNN, and two ImageNet-pretrained mobile architectures for ten-class EuroSAT RGB classification. The five reported final experiments use the saved 70/15/15 split and 64-by-64 inputs. The selected custom lightweight network uses 46,373 trainable parameters and 18.562 million convolution/linear multiply-accumulates (MACs) per image, compared with 94,762 parameters and 41.289 million MACs for the standard network. Recalculation from saved test predictions gives 93.90% and 94.37% accuracy, respectively. Model D uses 15,442 trainable parameters and 4.729 million MACs per image and reaches 93.63% test accuracy. MobileNetV2 reaches 98.12% with a recorded 9.180270 MB state dictionary; ShuffleNetV2 x0.5 reaches 96.37% with 3.231 million MACs. The results support different choices under different storage and accuracy constraints. They do not establish edge-device latency or robustness across random seeds. Historical CompactEuroSAT remains separate from the verified five-model comparison. The audit verifies saved metrics and source-derived costs, but cannot rerun trained-model inference without the excluded RGB images and trained checkpoints.

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1 Introduction

Low-resource image classification requires balancing recognition performance, model storage and computation. A high-accuracy model may exceed a device's storage or working-memory budget;
a small model may fit but lose useful class distinctions. The assignment therefore asks for a standard CNN (Model A), a depthwise-separable CNN with at most 100,000 trainable parameters (Model B), and two pretrained mobile architectures [4].
This report uses the code identifiers without rewriting experiment history. Model A is `model_a` (StandardCNN). The selected assignment Model B is `model_c` (WideLightweightCNN). The older `model_b` is an initial lightweight baseline, not the final report's Model B. CompactEuroSAT is a separate candidate, registered as `compact_eurosat`; it has not replaced `model_c`. The additional residual model is `model_d` (ResidualDepthwiseCNN).
The report distinguishes evidence levels. Predictions, confusion matrices and histories are available for the five reported final models and are checked independently. Saved weight-file sizes are recorded measurements in those reports, not new measurements in this cloud machine. Actual trained state dictionaries and RGB images are absent. Historical notebook outputs and Markdown-only claims provide weaker evidence and are labelled accordingly.

2 Dataset and preprocessing

2.1 Dataset, classes and resolution

EuroSAT contains Sentinel-2 land-cover imagery. Its RGB release is appropriate for the assignment's low-resolution setting and is distinct from CIFAR-10 [1].
The saved manifests describe 27,000 RGB images across ten classes. Image loaders require exactly $64 \times 64 \times 3$ RGB inputs rather than silently resizing incompatible files.
Raw image format and pixel hashes could not be rechecked during this audit because the images are not included in Git. The supplied dataset documentation and upstream preparation notebook identify the RGB source. The official dataset repository currently links Zenodo.

2.2 Saved stratified splits

The canonical files are `data/splits/train.csv`, `validation.csv` and `test.csv`. They contain 18,900, 4,050 and 4,050 images, respectively, implementing 70/15/15 stratification. No new split was saved. The audit checks class labels, class order, filenames, class-specific proportions, no overlapping paths, and 27,000 unique saved pixel hashes. This verifies the manifests; it does not independently verify absent image pixels.

Class	Train	Validation	Test
AnnualCrop	2100	450	450
Forest	2100	450	450
HerbaceousVegetation	2100	450	450
Highway	1750	375	375
Industrial	1750	375	375
Pasture	1400	300	300
PermanentCrop	1750	375	375
Residential	2100	450	450
River	1750	375	375
SeaLake	2100	450	450

Table 1: Counts recalculated from the canonical split manifests. The row order is the saved label order.

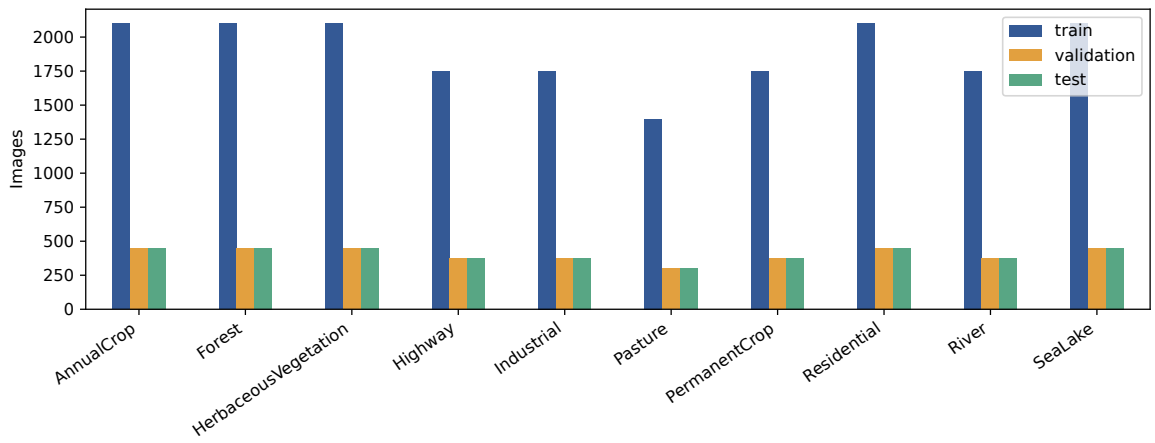


Figure 1: Class distribution. Class imbalance is mild but present, so macro metrics accompany overall accuracy.

2.3 Normalization and augmentation

Custom-run configurations save training RGB means
$(0.34390075, 0.37995413, 0.40754934)$ and standard deviations
$(0.20242461, 0.13668294, 0.11540833)$. These are reused without recalculating them on validation or test data. An RGB value is scaled to $[0, 1]$, then standardized per channel.
The common custom pipeline uses horizontal and vertical flips, each with probability 0.5, and a random quarter-turn. Evaluation uses no augmentation. MobileNetV2 and ShuffleNetV2 use
the same geometric augmentation and manifests, but the documented ImageNet means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$.
This deliberate normalization exception matches the pretrained backbones.

2.4 Preparation and provenance

The CLI's `prepare` command checks existing data and manifests; it does not download images, generate splits or estimate normalization. The optional `--check-images` hashes decoded pixels and checks RGB size. Both official download hosts tested during this audit returned proxy HTTP 403 responses. Supplying the original RGB files is required to run the new candidate. The assignment's external dataset sheet cannot be confirmed from this repository and remains a submission task. Appendix C records the cross-platform hash issue separately
from actual split membership.

3 Custom CNN architectures

3.1 Model A: standard convolution

Model A repeats a 3×3 standard convolution, BatchNorm, ReLU and 2×2 max pooling. The convolution widths are 32, 64 and 128. Convolutions use stride 1, padding 1 and no bias; pooling uses stride 2. Spatial resolution falls from 64 to 32, 16 and 8. Global average pooling produces 128 features and a biased 128-to-10 linear layer returns logits. There is no dropout or residual connection. Cross-entropy consumes the logits directly.

3.2 Model B: depthwise-separable convolution

The selected lightweight architecture uses three blocks with output widths 64, 128 and 256. Each block applies a 3×3 depthwise convolution (one spatial filter per input channel), then a 1×1 pointwise convolution, BatchNorm, ReLU and max pooling.
Both convolutions are bias-free; depthwise stride is 1 and padding is 1, while pointwise padding is 0.
Global average pooling and a biased 256-to-10 classifier complete the network.
There are no residual connections or dropout in this selected model.

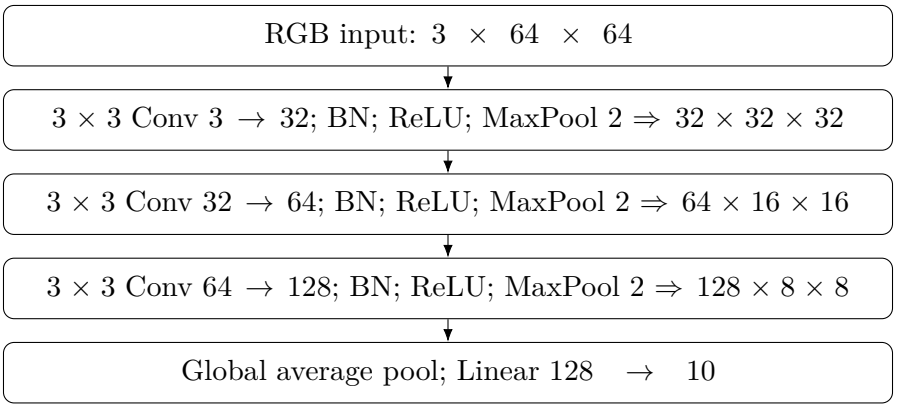


Figure 2: Model A, drawn from the executable architecture.

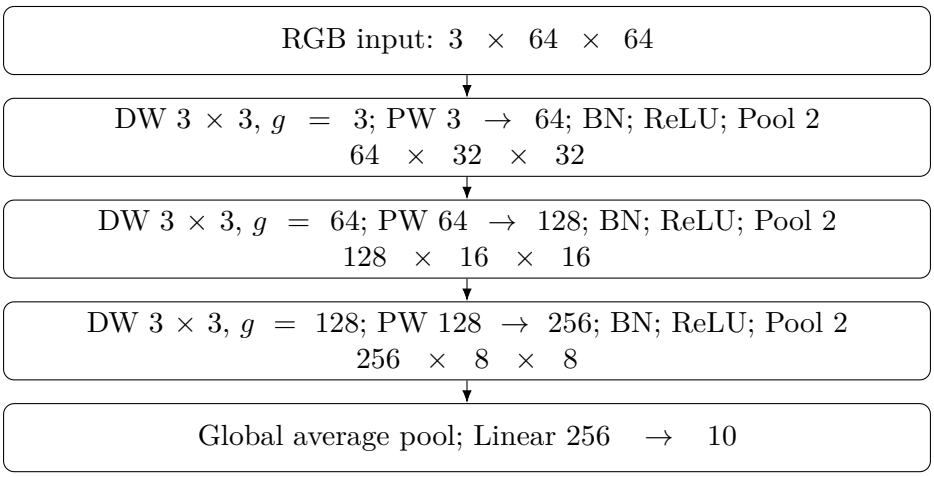


Figure 3: Selected Model B (code `model_c`). DW denotes depthwise and PW denotes pointwise convolution.

3.3 Parameter calculations

For a convolution with groups g, kernel K and optional bias, the parameter count is $C_{out}(C_{in}/g)K^2 + bC_{out}$, where $b = 1$ if a bias is present.
Depthwise convolution has $g = C_{in}$; pointwise convolution has $K = 1$.
BatchNorm contributes two trainable values per output channel. Running statistics are buffers, not trainable parameters. A linear layer contributes $n_{in}n_{out} + n_{out}$ with a bias.
For Model A the convolutions contribute $864 + 18,432 + 73,728 = 93,024$ parameters; BatchNorm adds $64 + 128 + 256 = 448$; the classifier adds $1,280 + 10 = 1,290$.
The total is $94,762$. For selected Model B the depthwise convolutions contribute $27 + 576 + 1,152 = 1,755$, pointwise convolutions contribute $192 + 8,192 + 32,768 = 41,152$, BatchNorm adds $128 + 256 + 512 = 896$, and the classifier adds $2,560 + 10 = 2,570$.
The total is $46,373 < 100,000$. Appendix tables list each executed leaf layer and check formula totals against actual parameter tensors.

3.4 Hardware-aware activation choice

ReLU computes $\max(0, x)$ and requires a comparison rather than an exponential.
It is therefore a reasonable low-cost activation for the custom networks. In-place ReLU can reuse intermediate storage when autograd permits it; this alone does not establish a peak-RAM budget. Global average pooling avoids a large flattened fully connected head.
BatchNorm may be fused into preceding convolutions for deployment, but such fusion, quantization and target-device execution were not measured here.

3.5 Computational cost

For one input, convolution MACs are

$$\text{MACs} = H_{out}W_{out}C_{out}(C_{in}/g)K^2. \tag{1}$$

Linear MACs are $n_{in}n_{out}$. Forward hooks independently count 41,288,960 MACs for Model A and 18,561,536 for selected Model B at $3 \times 64 \times 64$.
These counts omit bias additions, BatchNorm, activation, pooling, residual additions, memory access and data loading. They are not latency measurements.
For equal widths, separable convolution replaces $C_{in}C_{out}K^2$ weights with $C_{in}K^2 + C_{in}C_{out}$, but the selected model also changes widths, so the whole-network comparison is not a matched test of convolution type alone.

4 Optimizer selection and tuning

4.1 Methods and selection rule

The saved experiments compare Adam at 0.001, plain SGD at 0.003 and 0.01, and SGD with momentum 0.9 at 0.01, for the three custom architectures included in the optimizer sweep. All run for 30 epochs.

Within each run, the checkpoint has minimum validation loss; across configurations of the same architecture, minimum checkpoint validation loss determines the preferred configuration.

Test metrics are excluded from this rule.

Plain SGD follows a noisy negative gradient. Momentum accumulates a velocity, commonly $v_t = \mu v_{t-1} + g_t$ and $\theta_{t+1} = \theta_t - \eta v_t$; $\mu = 0.9$ retains more direction than $\mu = 0$. It can accelerate consistent descent and reduce oscillation, but may overshoot or require a different learning rate. Adam uses bias-corrected first and second gradient moments to adapt the update scale [2]. The recorded Adam learning rate is lower than the momentum-SGD rate, so differences cannot be attributed to momentum alone across all methods.

4.2 Observed experiments

Code	Optimizer	LR	Mom.	Epoch	Val. loss	Val. acc. (%)	Train s
model_a	adam	0.001	0	30	0.1721	94.12	26.78
model_a	sgd	0.003	0	29	0.4595	83.63	28.10
model_a	sgd	0.01	0	27	0.3796	87.21	26.01
model_a	sgd_momentum	0.01	0.9	30	0.1745	94.20	23.94
model_b	adam	0.001	0	27	0.2445	91.56	27.79
model_b	sgd	0.003	0	28	0.6512	78.30	27.93
model_b	sgd	0.01	0	28	0.4831	83.53	25.39
model_b	sgd_momentum	0.01	0.9	25	0.2347	91.83	26.10
model_c	adam	0.001	0	28	0.1776	94.00	30.53
model_c	sgd	0.003	0	29	0.5881	79.60	24.33
model_c	sgd	0.01	0	29	0.4166	85.01	34.29
model_c	sgd_momentum	0.01	0.9	29	0.2183	92.32	25.92

Table 2: Each row uses the minimum-validation-loss epoch. Train s is mean training-pass time over all 30 epochs; validation time is excluded. Historical code labels are retained.

Adam 0.001 gives the lowest recorded validation loss for Model A and `model_c`.

For the initial `model_b`, momentum SGD 0.01 gives the lowest loss.

Momentum SGD substantially improves on plain SGD 0.01 within this 30-epoch budget.

For Model A, momentum SGD has slightly higher selected validation accuracy than Adam, but Adam has lower validation loss, so Adam remains selected. Model B's wider channels recover validation performance relative to the initial lightweight baseline while staying below the parameter cap. One seed and a small learning-rate grid do not establish a universal optimizer ranking; neither optimizer family was exhaustively tuned.

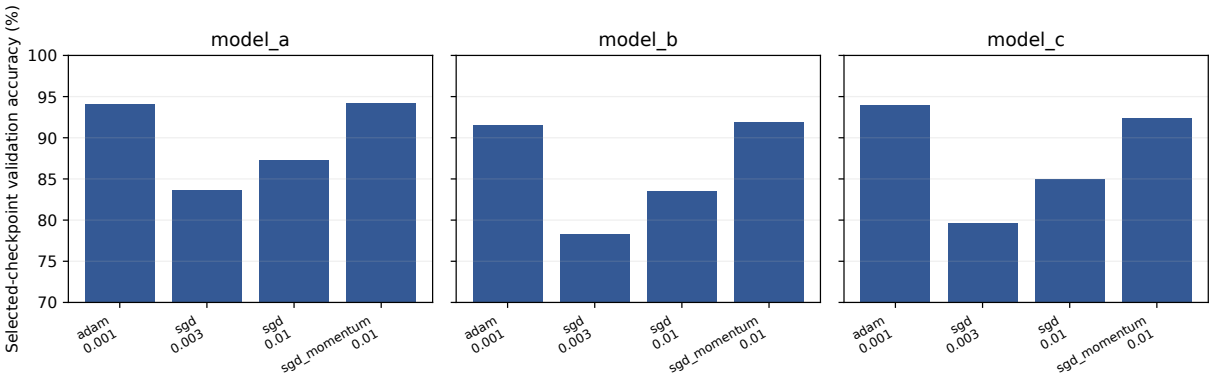


Figure 4: Validation accuracy at the selected epoch. The earliest lightweight baseline is code `model_b`; it is not the final selected Model B.

5 Custom model training and evaluation

5.1 Training setup and learning curves

The final custom runs use seed 42, batch size 64, Adam 0.001, weight decay 0.0001 and 30 epochs, satisfying the minimum-20-epoch requirement. The pipeline is FP32 cross-entropy with no learning-rate scheduler or early stopping. The selected epochs are 30 for Model A and 28 for selected Model B. Saved configurations identify CUDA; historical training timings come from the recorded machine, not this CPU cloud environment.

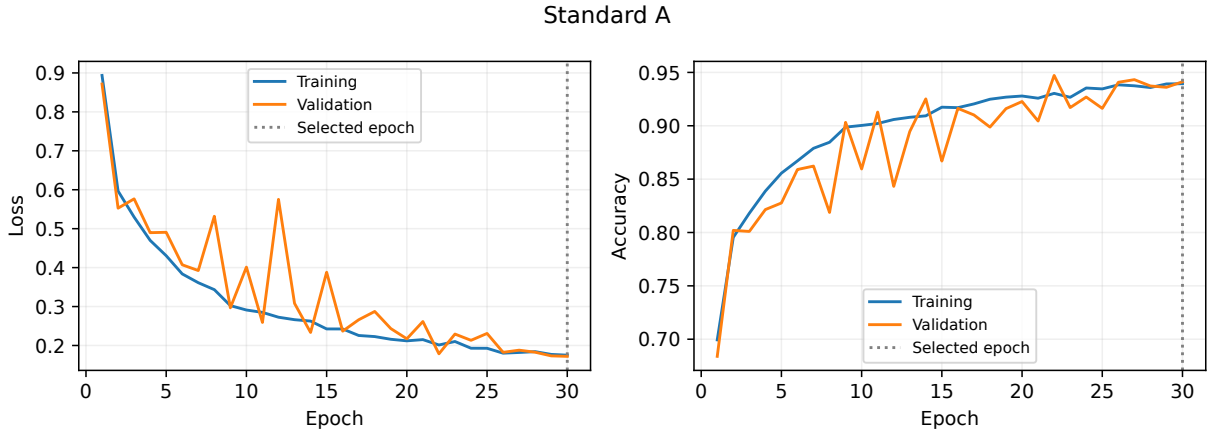


Figure 5: Model A training/validation history. The dotted line identifies the minimum-validation-loss epoch.

5.2 Test metrics and storage

The five saved prediction files contain the same 4,050 test filenames and true labels in the same order. Confusion matrices, precision, recall, F1 and overall accuracy were recomputed from these files and matched the saved reports within 10^{-12}.

This is an audit of saved predictions, not new test inference from trained checkpoints.

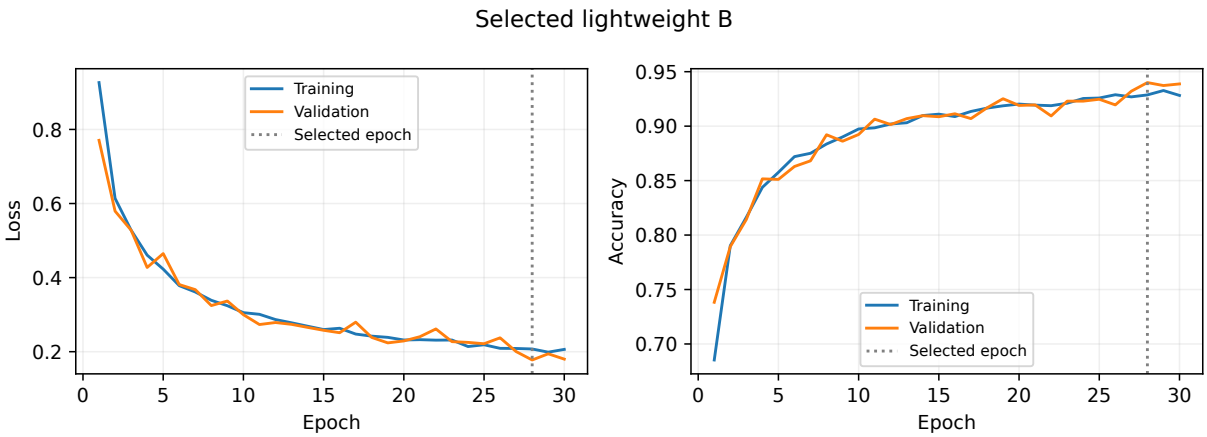


Figure 6: Selected Model B training/validation history. Curves are raw saved values, not smoothed estimates.

Model	Params	FP32 KB	Weights KB*	Train s	Test %
Standard A	94,762	379.048	387.218	26.78	93.90
Selected lightweight B	46,373	185.492	196.521	30.53	94.37

Table 3: Custom-model comparison. KB is 1,000 bytes. FP32 KB counts trainable parameter tensors only. *Weights KB is the historical serialized-file measurement recorded in metrics; the trained files are absent for remeasurement.

Selected Model B's test accuracy exceeds Model A's by approximately 0.47 percentage points (19 more correct predictions). This is a descriptive result on one split, not evidence of statistically robust superiority. The selected lightweight model halves parameter storage approximately and reduces convolution/linear computation, but its mean recorded training time is not lower. Implementation, kernel utilization, startup, data loading and shared-machine conditions can dominate wall-clock measurements.

5.3 Confusion matrices and class performance

Macro precision and recall weight the ten classes equally. Overall accuracy weights images; it can conceal lower recall for smaller classes. The full per-class tables in the appendix avoid treating overall accuracy as a complete error characterization. Confusions among agricultural/vegetation classes are plausible at this spatial resolution, but these matrices do not establish which visual features caused individual errors.

6 Lightweight pretrained architectures

6.1 MobileNetV2 and ShuffleNetV2 x0.5

MobileNetV2 uses inverted residual blocks with expansion, depthwise filtering and linear bottlenecks [6]. ShuffleNetV2 uses channel split/shuffle and efficient branches;

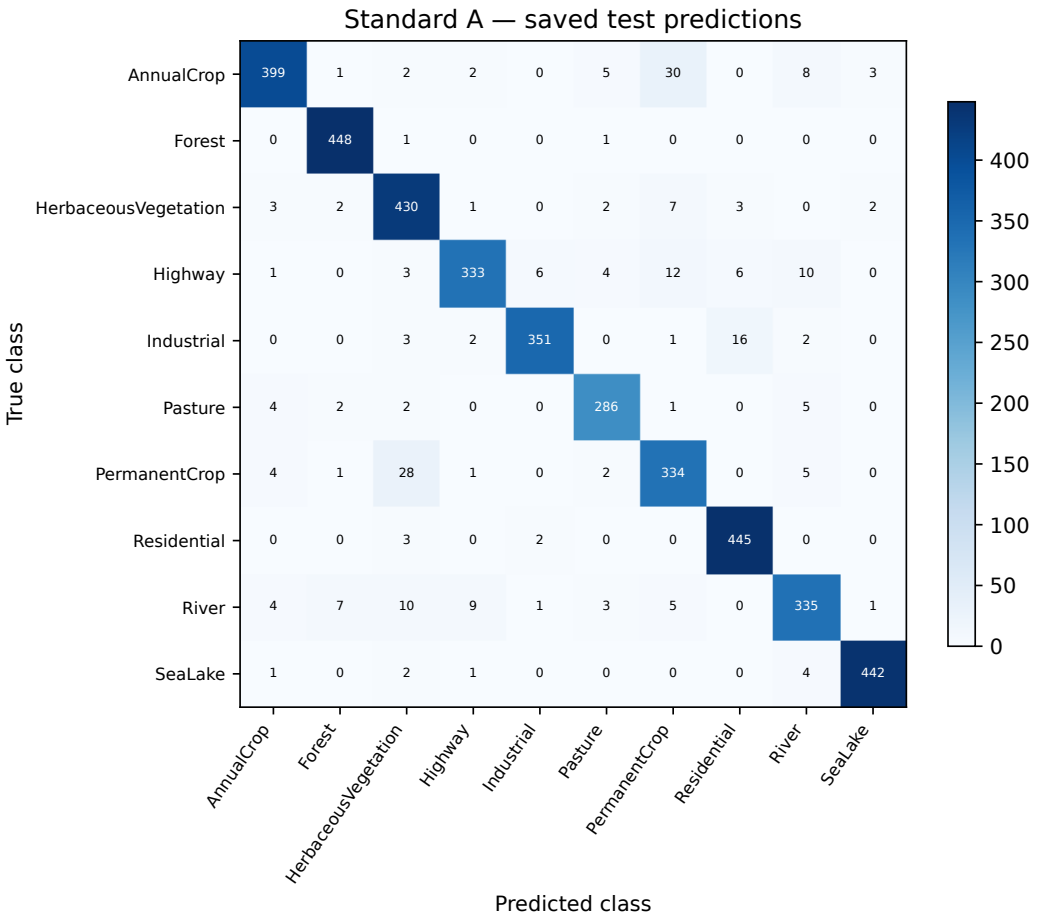


Figure 7: Model A test confusion matrix recalculated from saved predictions. Rows are true classes, columns predicted classes.

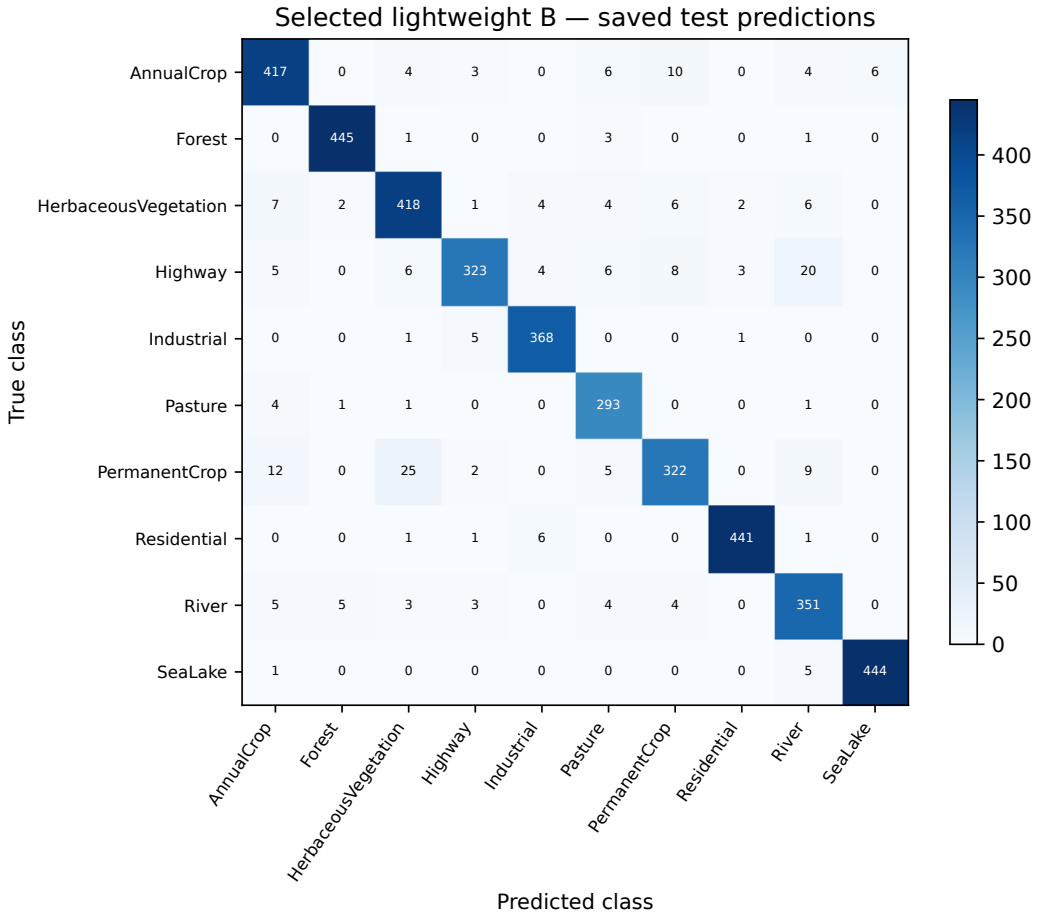


Figure 8: Selected Model B test confusion matrix recalculated from saved predictions.

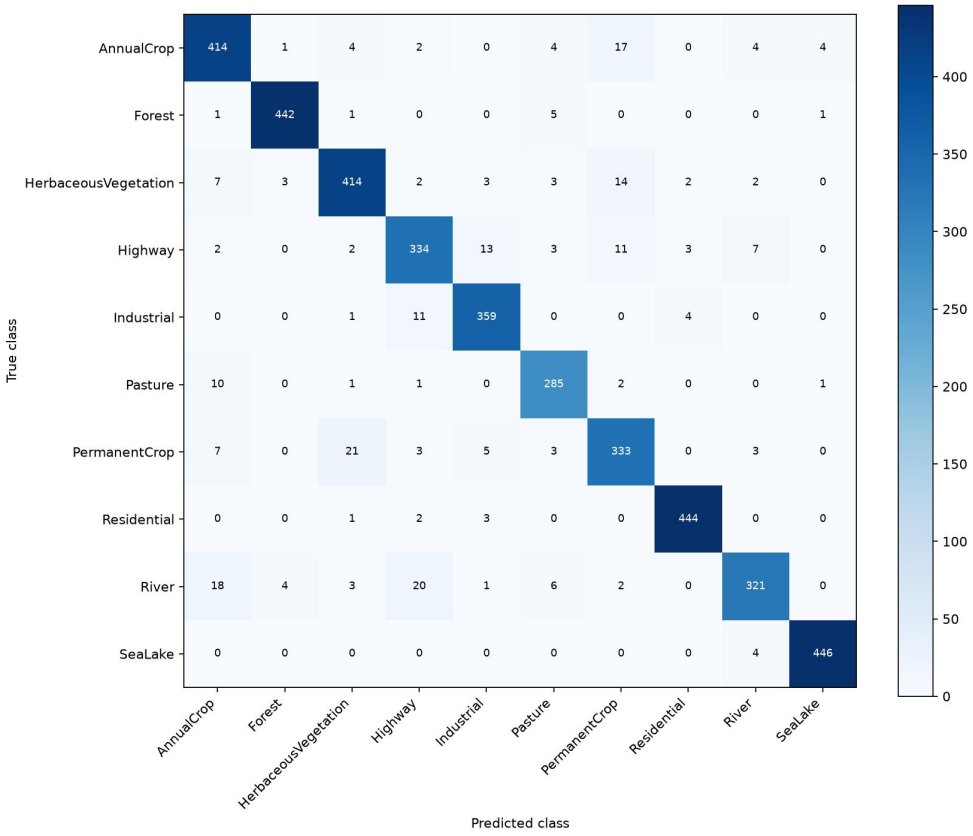


Figure 9: Model D test confusion matrix recalculated from saved predictions.

its design also considers practical execution cost beyond arithmetic counts [3].
The x0.5 variant has fewer channels than larger ShuffleNetV2 versions.
The executable loader first builds the original 1,000-class model, loads the official ImageNet state, and only then replaces the classifier with ten outputs. All layers remain trainable. MobileNetV2 uses `MobileNet_V2_Weights.IMAGENET1K_V2`; ShuffleNetV2 uses `ShuffleNet_V2_X0_5_Weights.IMAGENET1K_V1` [5].
Source-file SHA-256 prefixes and strict state dictionaries are checked, including cached/manual sources. Loading failure raises an error instead of falling back to random initialization.
During this audit, genuine official sources were hash-verified and synthetic tests checked backbone initialization and updates. Historical run configs also record successful pretrained initialization and full source hashes. This supports the intended training provenance; without the historical trained checkpoints, the exact final weight trajectories cannot be reverified.

6.2 Fine-tuning protocol and results

Both fine-tuning runs use the same saved split fingerprints, batch size 64, seed 42, 30 epochs, Adam 0.0001 and weight decay 0.0001. Minimum validation loss selects epochs 27 (MobileNetV2) and 29 (ShuffleNetV2). Inputs stay at 64-by-64 rather than the larger default ImageNet evaluation size. ImageNet normalization is the documented exception

to the custom preprocessing. No claim of matching published ImageNet benchmarks is made.

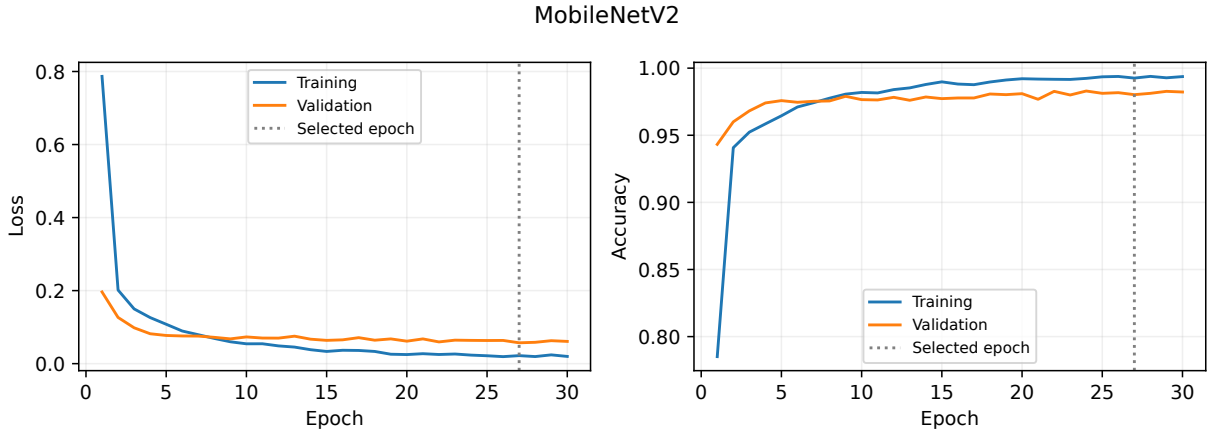


Figure 10: MobileNetV2 saved fine-tuning history.

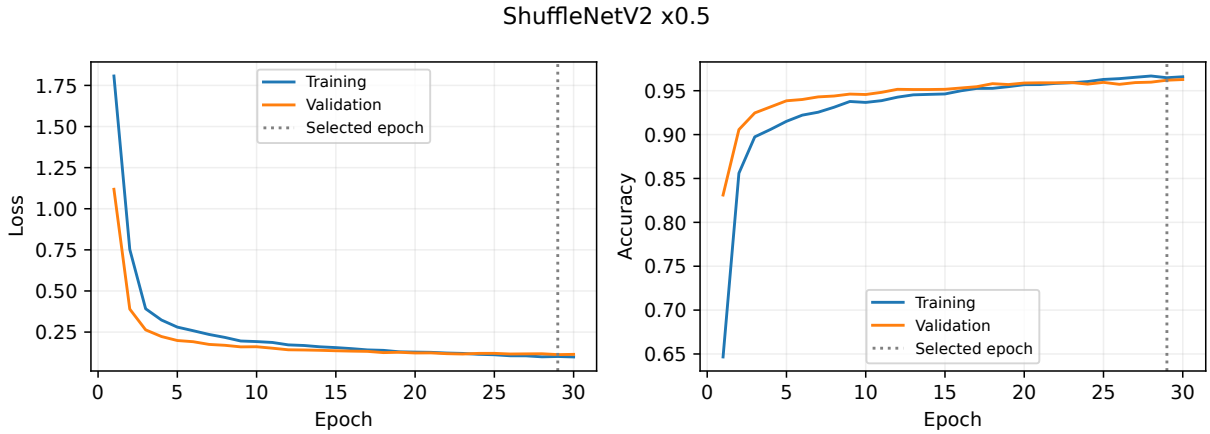


Figure 11: ShuffleNetV2 x0.5 saved fine-tuning history.

7 Final comparison and trade-offs

Model	Params	Weights MB*	MACs M	Val. %	Test %	Prec. %	Recall %	F1 %	Train s
Standard A	94,762	0.387218	41.289	94.12	93.90	93.90	93.70	93.75	26.78
Selected lightweight B	46,373	0.196521	18.562	94.00	94.37	94.23	94.25	94.18	30.53
Model D	15,442	0.082062	4.729	94.17	93.63	93.43	93.42	93.40	21.23
MobileNetV2	2,236,682	9.180270	24.461	98.02	98.12	98.11	97.95	98.02	42.16
ShuffleNetV2 x0.5	352,042	1.542078	3.231	96.20	96.37	96.21	96.23	96.21	37.26

Table 4: Verified saved-prediction comparison and source-derived costs. Precision, recall and F1 are macro averages. Validation accuracy is at minimum validation loss. *Weight size is recorded, not remeasured. Mean train seconds excludes validation but includes the first epoch.

7.1 Accuracy

MobileNetV2 has the highest saved test accuracy (98.12%), followed by ShuffleNetV2 x0.5 (96.37%), selected Model B (94.37%), Model A (93.90%) and Model D (93.63%).

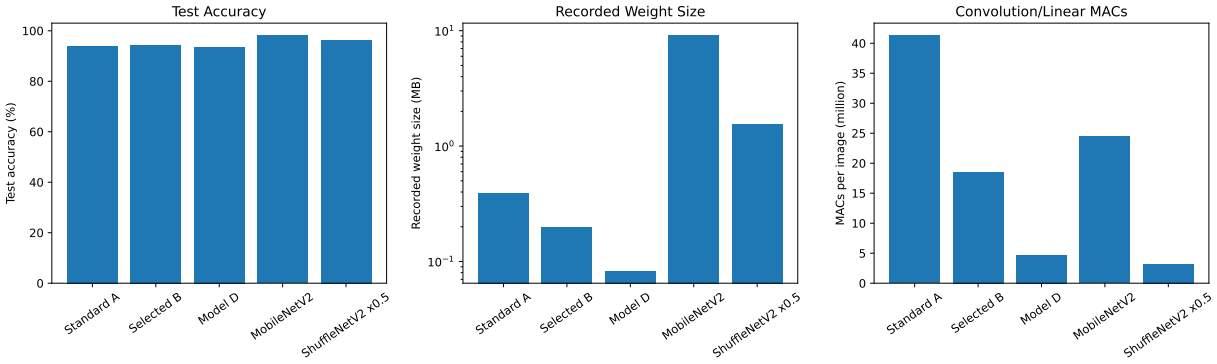


Figure 12: Accuracy, recorded serialized storage (log scale) and convolution/linear computation. These are complementary criteria; no panel measures edge latency.

Model D is an additional verified experiment rather than a replacement for the selected Model B. It is particularly relevant to the resource-constrained objective. Compared with selected Model B, it reduces trainable parameters from 46,373 to 15,442, a reduction of approximately 66.7%, and reduces counted convolution/linear MACs from 18.562 million to 4.729 million, a reduction of approximately 74.5%. The test-accuracy difference is 0.74 percentage points (94.37% versus 93.63%).

These results describe the recorded experiments and do not establish a causal advantage for residual connections because no matched residual/non-residual ablation was performed. All figures describe a single split and seed; small differences should not be generalized as stable rankings without repeated runs.

7.2 Memory footprint and computation

Among the four required models, selected Model B has the smallest parameter count and recorded weight file: 46,373 parameters and 0.196521 MB. Model D is smaller still, with 15,442 parameters and a recorded 0.082062 MB state-dictionary file, but it is an additional architecture rather than one of the required models. MobileNetV2 uses approximately 48.2 times as many parameters as Model B; ShuffleNetV2 uses 7.6 times as many. Serialized files include BatchNorm buffers and serialization overhead in addition to parameter tensors; training resume files also include optimizer and random-generator state. None of these sizes is peak inference RAM.

ShuffleNetV2 has the lowest counted computation among the five compared models: 3.231 million MACs/image, compared with 4.729 million for Model D and 18.562 million for selected Model B. Thus the smallest parameter model is not necessarily the lowest-compute model. Channel widths, feature resolutions and layer arrangement matter. Memory traffic, operator fusion and hardware kernels also affect latency, so a MAC ranking cannot replace a device benchmark.

7.3 Training time and deployment implications

Reporting means with and without the first epoch exposes startup sensitivity; it does not make timings comparable to an edge target. Under a strict storage limit, the custom selected Model B is attractive. When accuracy dominates and memory permits, MobileNetV2 is attractive. ShuffleNetV2 offers the lowest counted computation with intermediate storage. A final device choice still requires target latency, peak RAM, energy, quantization calibration and robustness measurements. These were not measured here.

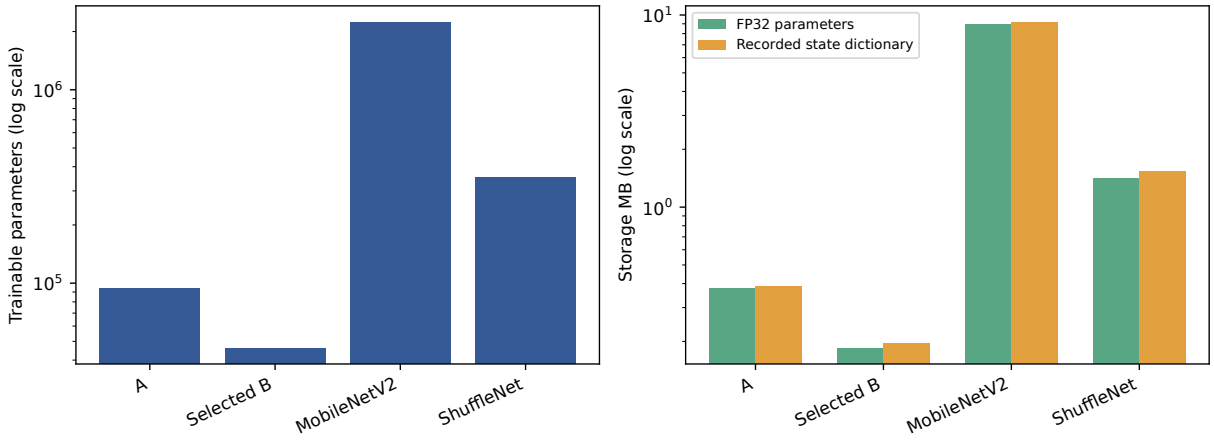


Figure 13: Parameters and storage. The raw FP32 parameter estimate and the recorded serialized state-dictionary size describe different storage scopes. Neither is peak RAM.

Model	Mean s	Mean excl. first s	Test pass s
Standard A	26.777	25.267	42.741
Selected lightweight B	30.532	25.624	3.463
MobileNetV2	42.158	36.505	4.113
ShuffleNetV2 x0.5	37.264	31.275	3.711

Table 5: Recorded timings, in seconds. Test pass time includes loading, transfers, loss and prediction collection, with CUDA synchronization; it is not model-only latency.

8 Additional architecture audit

8.1 CompactEuroSAT candidate

The `simple_model` branch contains a compact CNN with a standard 3×3 stem ($3 \to 16$), then depthwise-separable blocks $16 \to 32$ (stride 2), $32 \to 48$ (stride 2), $48 \to 48$ (stride 1, identity residual) and $48 \to 80$ (stride 2).

Each depthwise stage uses a 3×3 kernel, padding 1, no bias and channel-wise grouping; pointwise stages use 1×1 kernels and no bias. Both stages have BatchNorm, with ReLU after depthwise normalization and after the block. Global average pooling, dropout 0.10 and a biased 80-to-10 linear layer form the head. Source-derived cost is 11,466 parameters, 45,864 FP32 parameter bytes, and 3,882,784 convolution/linear MACs per image.

The historical notebook prints 94.716% test accuracy and approximately 94.64% validation accuracy. These are saved notebook claims, not a new verified controlled result. Reconstruction of its sorted-folder, stratified seed-42 split exactly matches the canonical memberships; the actual historical manifest files and trained checkpoint are not committed, so the reconstruction does not prove what files existed on the Kaggle machine.

Its optimizer protocol differs: batch size 128, maximum 60 epochs, patience 12, cosine scheduling, colour jitter, gradient clipping, CUDA mixed precision, no Adam weight decay argument, and checkpoint selection by validation macro-F1. Consequently its higher

printed accuracy is not used to select a replacement Model B.

The historical notebook and duplicate `lightweight_CNN` file remain unchanged.

A shared-CLI implementation is tested for exact evaluation-output equivalence after state loading, forward shapes and a real synthetic optimizer step. A controlled new run is prepared with the common manifests, 30 epochs, seed 42, batch 64, Adam 0.001, weight decay 0.0001, shared augmentation/normalization and minimum-validation-loss selection. Missing RGB files and blocked official download routes prevented execution. Its controlled validation/test metrics, timing and actual trained-file size are therefore **not available**.

The candidate's resource costs are promising, but no recommendation to replace selected Model B is justified before validation under the controlled protocol. Repeated selection against known test results would contaminate the test set; any further search should use validation data and ideally a genuinely held-out confirmation dataset.

8.2 ResidualDepthwiseCNN (Model D)

Branch `nipun` adds a 24-channel standard stem and residual depthwise blocks $24 \to 24$, $24 \to 48$ and $48 \to 96$, each downsampling by stride 2, followed by global average pooling and a 96-to-10 classifier. The completed experiment contains the saved configuration, 30-epoch training history, validation evidence and final-test evaluation artifacts.

The model has 15,442 trainable parameters and 4,728,768 convolution/linear MACs per input. The saved experiment uses Adam with learning rate 0.001, batch size 64, seed 42, 30 epochs and minimum validation loss for checkpoint selection. Epoch 30 has validation accuracy 94.17%, and the selected checkpoint achieves 93.63% test accuracy, 93.43% macro precision, 93.42% macro recall and 93.40% macro F1 on 4,050 test images.

The final-test evidence includes saved predictions, per-class metrics, confusion matrices and the recorded configuration/history. The recorded state-dictionary size is 0.082062 MB; the trained checkpoint itself is not committed to Git. These artifacts are sufficient to verify the reported prediction metrics and source-derived model costs, but not to reproduce trained-model inference without the original checkpoint and RGB images.

Model D is included in the verified model comparison as an additional architecture. However, no matched residual/non-residual ablation was performed, so the results do not establish that the residual connections themselves caused any performance difference.

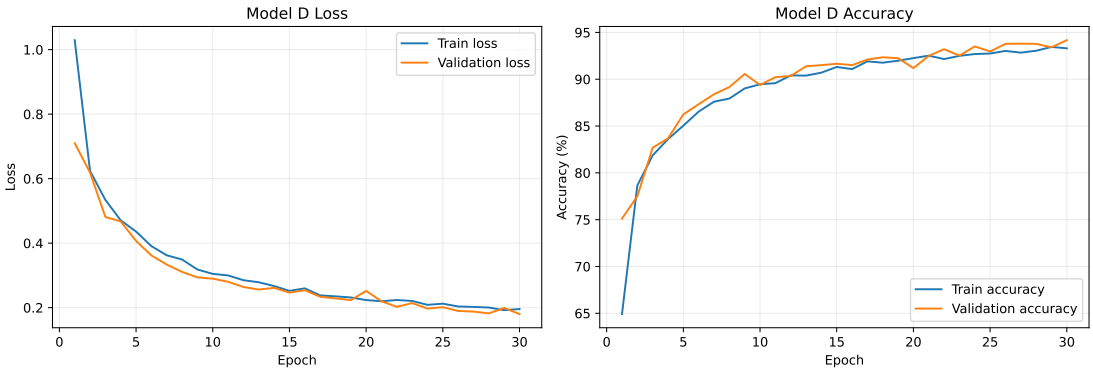


Figure 14: Model D training and validation history over the 30 recorded epochs.

9 Conclusion

The selected custom Model B meets the 100,000-parameter cap and provides a strong balance between accuracy and computational cost. Model D demonstrates that an even smaller residual depthwise architecture can be constructed with only 15,442 trainable parameters and 4.729 million convolution/linear MACs per image while achieving 93.63% test accuracy.

The verified comparison highlights complementary strengths. MobileNetV2 provides the highest recorded accuracy at 98.12%, ShuffleNetV2 x0.5 provides the lowest counted convolution/linear computation at 3.231 million MACs per image, and Model D provides the smallest parameter count and recorded state-dictionary size among the compared models. The selected Model B remains the preferred custom model because it achieves 94.37% accuracy while remaining below the 100,000-parameter limit.

Model D is an additional verified experiment rather than a causal residual-network ablation. No matched residual/non-residual experiment was performed, so the contribution of the residual connections themselves cannot be isolated. CompactEuroSAT remains a historical additional experiment because its 94.716% result was obtained under a different training protocol and is therefore not used to select or replace the controlled Model B. The conclusions are based on single recorded seeds and do not establish target-device latency, peak RAM, energy consumption, or robustness across repeated runs.

References

[1] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2019. URL `https://github.com/phelber/EuroSAT`.

[2] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. URL `https://arxiv.org/abs/1412.6980`.

[3] Ningning Ma, Xiangyu Zhang, Hai-Tao Zheng, and Jian Sun. ShuffleNet V2: Practical guidelines for efficient cnn architecture design. In *Proceedings of the European Conference on Computer Vision*, 2018. URL `https://arxiv.org/abs/1807.11164`.

[4] Sampath K. Perera. EN3150 Assignment 03: Simple convolutional neural network to perform classification, September 2026. Course assignment specification, supplied as EN3150_Assignment_03_In23.pdf in this repository.

[5] PyTorch Contributors. Torchvision 0.20: Models and pre-trained weights. Official software documentation, 2024. URL `https://docs.pytorch.org/vision/0.20/models.html`. Version 0.20 model and weight documentation.

[6] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018. URL `https://arxiv.org/abs/1801.04381`.

A Detailed layer calculations

A.1 Model A

Operation	Output C x H x W	Formula	Params
Conv2d k3 s1 p1 g1	32 x 64 x 64	32 x 3 x 3 x 3 + 0	864
BatchNorm2d	32 x 64 x 64	2 x 32	64
ReLU	32 x 64 x 64	0	0
MaxPool2d	32 x 32 x 32	0	0
Conv2d k3 s1 p1 g1	64 x 32 x 32	64 x 32 x 3 x 3 + 0	18,432
BatchNorm2d	64 x 32 x 32	2 x 64	128
ReLU	64 x 32 x 32	0	0
MaxPool2d	64 x 16 x 16	0	0
Conv2d k3 s1 p1 g1	128 x 16 x 16	128 x 64 x 3 x 3 + 0	73,728
BatchNorm2d	128 x 16 x 16	2 x 128	256
ReLU	128 x 16 x 16	0	0
MaxPool2d	128 x 8 x 8	0	0
AdaptiveAvgPool2d	128 x 1 x 1	0	0
Linear	10	128 x 10 + 10	1,290

Table 6: Executed leaf layers of Model A. Group g, stride s, padding p and kernel k are explicit. Outputs exclude the batch dimension.

A.2 Selected Model B

Operation	Output C x H x W	Formula	Params
Conv2d k3 s1 p1 g3	3 x 64 x 64	3 x 1 x 3 x 3 + 0	27
Conv2d k1 s1 p0 g1	64 x 64 x 64	64 x 3 x 1 x 1 + 0	192
BatchNorm2d	64 x 64 x 64	2 x 64	128
ReLU	64 x 64 x 64	0	0
MaxPool2d	64 x 32 x 32	0	0
Conv2d k3 s1 p1 g64	64 x 32 x 32	64 x 1 x 3 x 3 + 0	576
Conv2d k1 s1 p0 g1	128 x 32 x 32	128 x 64 x 1 x 1 + 0	8,192
BatchNorm2d	128 x 32 x 32	2 x 128	256
ReLU	128 x 32 x 32	0	0
MaxPool2d	128 x 16 x 16	0	0
Conv2d k3 s1 p1 g128	128 x 16 x 16	128 x 1 x 3 x 3 + 0	1,152
Conv2d k1 s1 p0 g1	256 x 16 x 16	256 x 128 x 1 x 1 + 0	32,768
BatchNorm2d	256 x 16 x 16	2 x 256	512
ReLU	256 x 16 x 16	0	0
MaxPool2d	256 x 8 x 8	0	0
AdaptiveAvgPool2d	256 x 1 x 1	0	0
Linear	10	256 x 10 + 10	2,570

Table 7: Executed leaf layers of selected Model B. All zero-parameter pooling/activation stages are included.

A.3 Model D

Operation	Output C x H x W	Formula	Params
Conv2d k3 s1 p1 g1	24 x 64 x 64	24 x 3 x 3 x 3 + 0	648
BatchNorm2d	24 x 64 x 64	2 x 24	48
ReLU	24 x 64 x 64	0	0
Conv2d k3 s2 p1 g24	24 x 32 x 32	24 x 1 x 3 x 3 + 0	216
BatchNorm2d	24 x 32 x 32	2 x 24	48
ReLU	24 x 32 x 32	0	0
Conv2d k1 s1 p0 g1	24 x 32 x 32	24 x 24 x 1 x 1 + 0	576
BatchNorm2d	24 x 32 x 32	2 x 24	48
AvgPool2d	24 x 32 x 32	0	0
ReLU	24 x 32 x 32	0	0
Conv2d k3 s2 p1 g24	24 x 16 x 16	24 x 1 x 3 x 3 + 0	216
BatchNorm2d	24 x 16 x 16	2 x 24	48
ReLU	24 x 16 x 16	0	0
Conv2d k1 s1 p0 g1	48 x 16 x 16	48 x 24 x 1 x 1 + 0	1,152
BatchNorm2d	48 x 16 x 16	2 x 48	96
Conv2d k1 s2 p0 g1	48 x 16 x 16	48 x 24 x 1 x 1 + 0	1,152
BatchNorm2d	48 x 16 x 16	2 x 48	96
ReLU	48 x 16 x 16	0	0
Conv2d k3 s2 p1 g48	48 x 8 x 8	48 x 1 x 3 x 3 + 0	432
BatchNorm2d	48 x 8 x 8	2 x 48	96
ReLU	48 x 8 x 8	0	0
Conv2d k1 s1 p0 g1	96 x 8 x 8	96 x 48 x 1 x 1 + 0	4,608
BatchNorm2d	96 x 8 x 8	2 x 96	192
Conv2d k1 s2 p0 g1	96 x 8 x 8	96 x 48 x 1 x 1 + 0	4,608
BatchNorm2d	96 x 8 x 8	2 x 96	192
ReLU	96 x 8 x 8	0	0
AdaptiveAvgPool2d	96 x 1 x 1	0	0
Linear	10	96 x 10 + 10	970

Table 8: Executed leaf layers of Model D. The table includes the main residual block operations and shortcut layers used by the executable architecture.

B Per-class saved test metrics

Class	Precision	Recall	F1	Support
AnnualCrop	0.9591	0.8867	0.9215	450
Forest	0.9718	0.9956	0.9835	450
HerbaceousVegetation	0.8884	0.9556	0.9208	450
Highway	0.9542	0.8880	0.9199	375
Industrial	0.9750	0.9360	0.9551	375
Pasture	0.9439	0.9533	0.9486	300
PermanentCrop	0.8564	0.8907	0.8732	375
Residential	0.9468	0.9889	0.9674	450
River	0.9079	0.8933	0.9005	375
SeaLake	0.9866	0.9822	0.9844	450

Table 9: Model A.

Class	Precision	Recall	F1	Support
AnnualCrop	0.9246	0.9267	0.9256	450
Forest	0.9823	0.9889	0.9856	450
HerbaceousVegetation	0.9087	0.9289	0.9187	450
Highway	0.9556	0.8613	0.9060	375
Industrial	0.9634	0.9813	0.9723	375
Pasture	0.9128	0.9767	0.9436	300
PermanentCrop	0.9200	0.8587	0.8883	375
Residential	0.9866	0.9800	0.9833	450
River	0.8819	0.9360	0.9082	375
SeaLake	0.9867	0.9867	0.9867	450

Table 10: Selected Model B.

Class	Precision	Recall	F1	Support
AnnualCrop	0.9020	0.9200	0.9109	450
Forest	0.9822	0.9822	0.9822	450
HerbaceousVegetation	0.9241	0.9200	0.9220	450
Highway	0.8907	0.8907	0.8907	375
Industrial	0.9349	0.9573	0.9460	375
Pasture	0.9223	0.9500	0.9360	300
PermanentCrop	0.8786	0.8880	0.8833	375
Residential	0.9801	0.9867	0.9834	450
River	0.9413	0.8560	0.8966	375
SeaLake	0.9867	0.9911	0.9889	450

Table 11: Model D.

Class	Precision	Recall	F1	Support
AnnualCrop	0.9692	0.9800	0.9746	450
Forest	0.9868	0.9978	0.9923	450
HerbaceousVegetation	0.9758	0.9867	0.9812	450
Highway	0.9812	0.9733	0.9772	375
Industrial	0.9892	0.9733	0.9812	375
Pasture	0.9727	0.9500	0.9612	300
PermanentCrop	0.9781	0.9520	0.9649	375
Residential	0.9782	0.9978	0.9879	450
River	0.9866	0.9840	0.9853	375
SeaLake	0.9934	1.0000	0.9967	450

Table 12: MobileNetV2.

Class	Precision	Recall	F1	Support
AnnualCrop	0.9510	0.9489	0.9499	450
Forest	0.9781	0.9933	0.9857	450
HerbaceousVegetation	0.9600	0.9600	0.9600	450
Highway	0.9288	0.9387	0.9337	375
Industrial	0.9813	0.9813	0.9813	375
Pasture	0.9385	0.9667	0.9524	300
PermanentCrop	0.9322	0.9173	0.9247	375
Residential	0.9911	0.9933	0.9922	450
River	0.9667	0.9280	0.9469	375
SeaLake	0.9933	0.9956	0.9945	450

Table 13: ShuffleNetV2 x0.5.

C Reproducibility and audit limits

The artifact generator `report/generate.py` checks manifests, reconstructs compact split membership, recomputes all final classification metrics, checks per-class and confusion files, recalculates parameter/MAC counts, and creates figures directly from histories/predictions. `report/audit/report_data_sources.csv` maps generated values to their source files and evidence level. Model-layer CSVs include operation dimensions, formulas, parameter counts and MACs. Recorded file sizes are not silently relabelled as new measurements.

The canonical final reports for the four required models are in timestamped `outputs/reports/evaluate-test-*` directories. Model D's final-test evidence is stored under `outputs/reports/model_d_numpy126_final_test/`, with its training configuration and history under `outputs/custom_cnn/model_d_residual_numpy126/`. The raw Git

CSV bytes use LF line endings, whereas the historical saved split and source hashes match CRLF serializations. This was verified in memory without altering files or expected hashes. Two final-audit test methods fail for this byte-level difference. The existing evaluation CLI also checks literal split fingerprints; replaying a Windows run on an LF checkout needs an explicit, audited portability change or a separate matching serialization, not a changed expected hash. Historical exact-resume additionally requires the recorded device, workers, Python and package versions and full optimizer/RNG checkpoints.

The source branch audit uses fetched remote refs. Dataset/setup/standard-CNN/integration branches

are already ancestors of main; `heshan` equals main. `simple_model` was 18 commits ahead

and 19 behind; it was synchronized with a normal merge after creating safety branches.

`nipun` was integrated into `main` after the Model D environment and final-test artifacts were corrected. The Model D implementation and its verified evidence are therefore part of the final main-branch state.

Repository history begins on 10 September 2026 and includes later dataset, model and integration commits; the visible history is preserved rather than manufactured for submission.

Requirement / reproducibility item	Audit outcome
Assignment PDF read	Supplied filename differs from request; contents read.
Dataset source, RGB, 64×64	Documented; loader enforces shape; raw images absent.
70/15/15 splits, saved filenames/class order	Verified from manifests; no overlapping paths/hashes.
Normalization and augmentation	Verified from source/configs; no statistics recalculated.
Seed, optimizer, LR, batch size, epochs	Saved configs/histories inspected; one seed each.
Checkpoint selection	Minimum validation loss in source/history; historical checkpoints absent.
Model A standard conv/max pooling	Verified from executable source.
Model B depthwise and parameter cap	46,373 parameters, independently verified.
Activation/hardware justification	ReLU explained; hardware speed not measured.
Optimizer comparisons and momentum	Twelve saved runs checked.
At least 20 custom training epochs	Thirty rows per saved run; no new retraining claim.
Test metrics and confusion matrices	Recomputed from five saved prediction files.
Model size and time comparison	Recorded sizes/timings labelled; files absent.
Two genuine pretrained mobile models	Official initialization tested; historical configs checked.
Accuracy/memory/computation discussion	Included; latency/RAM/energy unmeasured.
Controlled CompactEuroSAT evaluation	Blocked by missing images / download-route 403.
Model D final result verification	Configuration, 30-epoch history, predictions, per-class metrics and confusion matrices checked; 93.63% test result verified.
Dataset assignment sheet update	External sheet status unknown; group action required.

Requirement / reproducibility item	Audit outcome
Group number and names	Group 54 supplied by user; names agree in Git, spelling confirmation required.
GitHub/profile link and history	Included; original history preserved.
Report/code separate Moodle submission	Not performed; group must submit once.